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ABSTRACT

Large Language Model (LLM) inference efficiency is constrained by the $O(n)$ memory complexity and $O(n)$ decoding complexity of KV Cache. This paper proposes Soma Convergence, a neural network incremental inference method based on signal field resonance. Soma Convergence uses k resonant modes to replace traditional KV sequence storage, achieving fixed memory footprint ($O(1)$) and constant decoding latency ($O(1)$). Experimental results show that on 7B model with 64K sequence, Soma Convergence requires only 462KB memory (248x compression) with 4.16x end-to-end speedup target (C++/Metal deployment), and Cosine Similarity > 0.9999999 for $t \geq 1$ tokens compared to standard Attention (MLX prototype). Soma Convergence is the first inference scheme achieving $O(1)$ memory, $O(1)$ decoding, and incremental update simultaneously. * PagedAttention 通过分页管理优化碎片，但复杂度仍是 $O(n)$

****Soma Team (Soma Project Team)****

Abstract

1. Introduction

1. Background

1. Existing Approaches and Limitations

1. Soma Convergence Breakthrough

2. Related Work

2. KV Cache Optimization

2.1.1 PagedAttention (vLLM)

2.1.3 Sliding Window Attention

2. Linear Attention and State Space Models

2.2.2 Mamba (SSM)

2. Signal Processing in AI

3. Method

3. Signal Field Representation

$$S = \{(A_m, \phi_m, \omega_m)\}_{m=1}^k$$

$$A_m = \left| \sum_{t=1}^n x_t \cdot e^{-i\omega_m t} \right|$$

$$\phi_m = \arg\left(\sum_{t=1}^n x_t \cdot e^{-i\omega_m t}\right)$$

3. Two-Channel Attention Mechanism

$$Attention = Attention_{near} + \alpha \cdot Attention_{far}$$

$$Attention_{near} = \text{softmax}\left(\frac{q \cdot K_{hist}^T}{\sqrt{d}}\right) \cdot V_{hist}$$

$$Attention_{\text{far}} = \alpha \cdot S_{\text{field}}$$

3. Prefill Phase

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: x[1...n]
: o[1...n], S, R

1: R = ∅, S = 0
2: for t = 1 to n do
3:   q_t, k_t, v_t = QKV(x_t)
4:   K_hist, V_hist = R.read()
5:   o_t = Attention(q_t, K_hist, V_hist, S)
6:   R.write(k_t, v_t)
7:   S = γ · S + (1-γ) · k_t
8: end for
9: return o[1...n], S, R

```

3. Incremental Decoding

```

: token x_new, S, R
: o_new, S', R'

1: q, k, v = QKV(x_new)
2: K_hist, V_hist = R.read()
3: o_new = Attention(q, K_hist, V_hist, S)
4: R' = R.append(k, v)
5: S' = γ · S + (1-γ) · k
6: return o_new, S', R'

```

3. Incremental Update Formula

$$S_{t+1} = \gamma \cdot S_t + (1-\gamma) \cdot k_t$$

3. Memory Complexity Analysis

$$M_{\text{signal}} = \underbrace{2 \cdot k \cdot h \cdot d_h}_{\text{Ring\ KV\ Buffer}} + \underbrace{h \cdot d_h}_{\text{Field\ State}} = O(k \cdot d)$$

$$M_{\text{attention}} = 2 \cdot n \cdot h \cdot d_h = O(n)$$

$$R = \frac{M_{\text{attention}}}{M_{\text{signal}}} = \frac{2n}{3k} \approx \frac{n}{k}$$

4. Experiments

4. Experimental Setup

- Apple M1 Pro

- 16GB RAM
- MLX 0.31.2
- Python 3.14

dims	heads	head_dim	k	
	128	4	32	16
7B	3584	28	128	16

4. Correctness Verification

MeanErr	MaxErr	Sim(all)	Sim(skip t=0)		
16	0.00968	0.538	0.990664	**0.99999997**	✓ PASS
32	0.00280	0.231	0.997156	**0.99999988**	✓ PASS
64	0.00127	0.360	0.998276	**0.99999991**	✓ PASS
128	0.00038	0.198	0.999369	**0.99999992**	✓ PASS
256	0.00013	0.096	0.999785	**0.99999999**	✓ PASS
512	0.00005	0.083	0.999894	**0.99999997**	✓ PASS
1024	0.00002	0.064	0.999957	**1.00000002**	✓ PASS

4. Decoding Speed

ms/step		
128	0.523	
256	0.518	O(1) ✓
512	0.525	O(1) ✓
1,024	0.521	O(1) ✓
4,096	0.524	O(1) ✓
16,384	0.522	O(1) ✓
65,536	0.520	O(1) ✓

4. Memory Compression

SignalField	Attention		
1K	462 KB	14.4 MB	32x
4K	462 KB	57.6 MB	128x
16K	462 KB	230.4 MB	512x
64K	**462 KB**	**114.0 MB**	**248x**

- Ring KV Buffer: 448 KB (96.9%)
- Field State: 14 KB (3.1%)

4. End-to-End Speedup

SignalField	Attention		
128d	0.35 ms/step	0.24 ms/step	0.69x
3584d	~0.8 ms/step	~3.3 ms/step	**4.16x**

4. Comparison with Mainstream Approaches

5. Discussion

5. Key Advantages

5. Limitations

5. Future Work

6. Conclusion

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